Lavish Avinash

Software Engineer | Backend Systems Architect | AI & Agentic Systems Developer | Product Builder
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PROFESSIONAL SUMMARY

Software engineer completing a B.E. in Information Science and Engineering (2027), focused on backend systems, AI-agent infrastructure and product engineering. Backend and system architect for VMR Mangoes, a perishable-goods commerce platform, responsible for its entire backend: a Node.js, Express and PostgreSQL 16 modular monolith with FEFO batch allocation, time-limited stock holds, idempotent Razorpay payment handling, four-role access control, a transactional notification outbox and 900+ automated tests. Develops AMU, an MCP-based code impact analysis tool that gives AI coding agents source-backed dependency paths before they change code, and Aurora, a Transformer-based multi-SKU demand forecasting system. Winner of the DevDays Bengaluru Build OpenAI Hackathon and shortlisted for ISRO IRoC-U. Also produces brand and video content for an elevator manufacturer, bringing client communication and business context to engineering work.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL; familiar: Java, Dart
Backend: Node.js, Express.js, FastAPI, REST API design, Authentication & authorisation (JWT, RBAC, TOTP), Webhooks & idempotency
Databases: PostgreSQL, MySQL, SQLite, Schema design & migrations, Transactions & locking; familiar: MongoDB
Frontend: React, Next.js, HTML, CSS; familiar: Tailwind CSS, Flutter
AI / ML: PyTorch, Transformers (time series), Scikit-learn, Pandas, NumPy, Feature engineering, Model evaluation; familiar: OpenCV, YOLO
AI agents: Model Context Protocol (MCP), AI agent integrations, Dependency graph analysis
Infrastructure & tooling: Git, GitHub Actions (CI/CD), Vercel, Neon, Docker, Playwright, Linux, Postman; familiar: ROS, Jupyter
Analytics: Microsoft Excel; familiar: Power BI, Tableau

PROFESSIONAL EXPERIENCE

Backend Architect & System Architect Present
VMR Mangoes, Bengaluru, India

- Architected the complete VMR Mangoes backend as a modular monolith on Node.js, Express and PostgreSQL 16, serving the storefront API, staff operations API and Razorpay payments from one service and one database.
- Designed a transactional stock engine in which time-limited holds, payment confirmation and FEFO batch allocation commit in a single PostgreSQL transaction, using row-level locks (FOR UPDATE, SKIP LOCKED) to prevent overselling.
- Built idempotent payment handling: HMAC-SHA256 webhook verification with timing-safe comparison, replay-safe payment outcomes, idempotency keys on refunds and automatic refunds of duplicate or wrong-amount payments.
- Implemented four-role access control (manager, operator, packer, QC) with TOTP two-step sign-in for staff and an activity log recording every staff change.
- Wrote 51 versioned SQL migrations applied under a PostgreSQL advisory lock so concurrent deploys never race; set up GitHub Actions CI, deployment and backups for Vercel with Neon Postgres, plus a Docker setup.
- Covered the backend with 900+ automated test cases and the React admin panel with Playwright end-to-end, axe accessibility and API-contract tests.
- Modelled operations from farm batch to doorstep: QC by ripeness, pack lists, own-rider delivery manifests with 4-digit delivery codes, claims settled by replacement, preorders and bulk business orders.
- Added a transactional outbox for WhatsApp order updates (queued in the same transaction as each order step, retried with backoff) and Prometheus metrics, Sentry and structured logging for observability.

Software Engineering Intern Present
VMR Cloud Solutions, Bengaluru, India

Digital Content Producer & Brand Communications Associate Present
Yashoda Elevators, Bengaluru, India

- Conceptualise promotional campaigns and brand stories for the company's digital channels.
- Shoot corporate and promotional video with mobile filmmaking equipment, and edit and finish short-form social content in post-production.
- Support the brand's visual communication and online presence, working directly with the business on what each piece needs to say.

PROJECTS

AMU: Anti Messup Agent

Model Context Protocol (MCP), Dependency graphs, Static code analysis

- Built an AI-agent-oriented code impact analysis system that assesses the downstream effects of a proposed repository change before an agent makes it.
- Designed a repository dependency graph tracing direct and transitive relationships between functions and modules, exposed to AI coding agents as MCP tools with source-backed dependency paths.
- Surfaced related tests and documentation for targeted validation, with coverage and uncertainty information so agents can tell evidence from gaps.

Aurora

Python, PyTorch, FastAPI, Next.js 14, MySQL

- Architected an end-to-end multi-SKU demand forecasting pipeline around a Transformer time-series model in PyTorch (30-day input, 5 features, 7-day horizon).
- Turned forecasts into automated safety stock and reorder point calculations served through a FastAPI inference API and a Next.js 14 interface.

Autonomous UAV: ISRO IRoC-U

Python, ROS, Computer vision, Embedded systems, Sensors

- Designed and implemented ROS-based autonomous navigation and real-time obstacle detection modules for a UAV shortlisted in the ISRO IRoC-U challenge.
- Integrated onboard sensors with the flight controller and contributed to system architecture and safety and performance requirements.

Spotlizer

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

- Built a modular ML pipeline predicting song popularity from Spotify audio features: preprocessing, feature engineering and scaling, EDA and a Random Forest classifier.
- Evaluated with accuracy, precision, recall, F1-score and confusion matrices, and shipped batch prediction behind a Streamlit interface.

Snibby

Flutter, Dart, Riverpod, SQLite

- Built a Flutter desktop app that OCRs screenshots, classifies them (chats, payments, coding errors, notes, documents), generates tags, detects duplicates and builds collections, with metadata stored locally in SQLite.

EDUCATION

Bachelor of Engineering, Information Science and Engineering — Expected 2027

CMR Institute of Technology, Bengaluru

Class XII (Science) — 2023

SRV Boys Higher Secondary School, Rasipuram, Tamil Nadu

Class X (CBSE) — 2021

Vagdevi Vilas School, Bengaluru

AWARDS & ACHIEVEMENTS

- Winner, DevDays Bengaluru Build OpenAI Hackathon. Top 10 of approximately 1,000 participating teams.
- Shortlisted, ISRO Robotics Challenge (IRoC-U). Autonomous UAV navigation and obstacle detection.
- First Place, Flex It Fitness Competition. Cash prize for outstanding performance.
- First Place, inter-college mono-action competition

LEADERSHIP & ACTIVITIES

- Organised a SQL Hackathon for the Equal Opportunity Club and the Information Science Department (2025)
- Social Media Head, Innovation Club
- Active member, Peer Support Team (2023–2026)
- Member, Theatre Club (Take a Bow)
- Participant, Smart India Hackathon and UNLOX Gen AI Hackathon (2025)
- Assistant Scouts Captain (2022)
- Languages: English, Kannada, Hindi, Tamil